



Setting Up Prometheus: A Beginner's Guide

Introduction

What is Prometheus?

Prometheus is an **open-source monitoring and alerting toolkit** designed for **reliability and scalability** in cloud-native environments. It is widely used for **monitoring metrics, gathering time-series data, and generating alerts.**

Why Use Prometheus?

- **Powerful Data Model:** Stores time-series data efficiently.
- **Pull-Based Monitoring:** Fetches metrics from targets instead of waiting for them to be sent.
- **Highly Scalable:** Works well in cloud environments.
- **Integrated Alerting:** Built-in support for **Alertmanager**.
- **Flexible Querying:** Uses **PromQL** for data analysis.

1. Installing Prometheus

Prerequisites

- A **Linux-based system (Ubuntu/CentOS) or Docker.**
- Basic understanding of **Linux commands.**
- **Node Exporter** (if monitoring system metrics).

Step 1: Download and Install Prometheus

For Linux Users

```
# Update system packages
sudo apt update && sudo apt install wget -y
```



```
# Create a Prometheus user
```

```
sudo useradd --no-create-home --shell /bin/false prometheus
```

```
# Create directories for Prometheus
```

```
sudo mkdir /etc/prometheus /var/lib/prometheus
```

```
# Download Prometheus
```

```
wget
```

```
https://github.com/prometheus/prometheus/releases/latest/download/prometheus-linux-amd64.tar.gz
```

```
# Extract files
```

```
tar xvf prometheus-linux-amd64.tar.gz
```

```
# Move binaries to /usr/local/bin/
```

```
sudo mv prometheus promtool /usr/local/bin/
```

```
# Move config files
```

```
sudo mv consoles/ console_libraries/ prometheus.yml /etc/prometheus/
```

Step 2: Configure Prometheus

Edit the Prometheus configuration file (</etc/prometheus/prometheus.yml>):

```
global:
```

```
  scrape_interval: 15s
```



```
scrape_configs:
  - job_name: 'prometheus'
    static_configs:
      - targets: ['localhost:9090']
```

Step 3: Create a Systemd Service for Prometheus

```
sudo tee /etc/systemd/system/prometheus.service <<EOF
[Unit]
Description=Prometheus Monitoring System
After=network.target

[Service]
User=prometheus
ExecStart=/usr/local/bin/prometheus
--config.file=/etc/prometheus/prometheus.yml

[Install]
WantedBy=multi-user.target
EOF
```

Enable and start the Prometheus service:

```
sudo systemctl daemon-reload
sudo systemctl enable prometheus
sudo systemctl start prometheus
```



Step 4: Access the Prometheus Dashboard

- Open a browser and go to **`http://localhost:9090`**.
- Run queries using **PromQL**.

2. Integrating Prometheus with Node Exporter

Step 1: Install Node Exporter

```
wget
https://github.com/prometheus/node_exporter/releases/latest/download/node_e
xporter-linux-amd64.tar.gz

tar xvf node_exporter-linux-amd64.tar.gz
sudo mv node_exporter /usr/local/bin/
```

Step 2: Configure Node Exporter Service

```
sudo tee /etc/systemd/system/node_exporter.service <<EOF
[Unit]
Description=Node Exporter
After=network.target

[Service]
User=prometheus
ExecStart=/usr/local/bin/node_exporter

[Install]
WantedBy=default.target
```



EOF

Enable and start Node Exporter:

```
sudo systemctl daemon-reload
sudo systemctl enable node_exporter
sudo systemctl start node_exporter
```

Step 3: Add Node Exporter to Prometheus

Modify `/etc/prometheus/prometheus.yml`:

```
scrape_configs:
  - job_name: 'node_exporter'
    static_configs:
      - targets: ['localhost:9100']
```

Restart Prometheus:

```
sudo systemctl restart prometheus
```

3. Know More: FAQs

1. How does Prometheus collect metrics?

- Prometheus **pulls** metrics from configured endpoints at defined intervals.

2. What is PromQL?

- **Prometheus Query Language (PromQL)** is used for analyzing time-series data.

3. How do I view Prometheus metrics?

- Go to **`http://localhost:9090`** and use the **Graph** tab.



4. Can I use Prometheus with Grafana?

- Yes, Prometheus is a common **data source for Grafana** dashboards.

Conclusion

Prometheus is a **powerful and scalable monitoring solution**. By setting it up with **Node Exporter**, you can **monitor system performance, create alerts, and integrate it with Grafana** for advanced visualization.